

DOERNICK, Germany

WMO: 101500

Lat: 54.1656N

Lon: 10.3519E

Elev: 90

StdP: 14.65

Time Zone: 1.00 (EUC)

Period: 01-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	14.6	20.0	11.0	9.7	16.8	15.4	12.0	21.7	28.7	43.7	25.2	43.1	4.4	140	0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.6	80.7	66.8	77.3	65.6	74.2	64.1	68.8	77.2	67.0	74.4	65.5	71.9	7.3	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.9	95.8	71.5	64.3	90.5	70.0	62.9	86.2	68.7	33.0	77.1	31.6	74.8	30.4	71.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.7	18.8	16.7	DB	7.7	87.5	10.3	3.0	0.3	89.6	-5.7	91.4	-11.5	93.0	-19.0	95.2
			WB	7.3	72.5	10.2	1.3	-0.1	73.5	-6.1	74.3	-11.8	75.0	-19.3	76.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.0	34.1	33.9	38.3	46.7	53.2	58.6	63.5	63.0	57.5	49.3	42.2	35.2
	DBStd	12.39	8.28	7.12	7.31	6.05	5.47	5.10	4.93	4.51	4.85	5.97	6.06	7.38
	HDD50	2263	493	452	366	133	29	2	0	0	2	86	240	460
	HDD65	6287	958	871	829	548	368	204	91	94	228	486	685	925
	CDD50	1544	0	1	2	35	128	259	418	402	229	65	5	0
	CDD65	91	0	0	0	0	1	10	43	32	5	0	0	0
	CDH74	701	0	0	0	0	14	82	391	185	28	1	0	0
	CDH80	113	0	0	0	0	2	5	77	28	1	0	0	0
Wind	WSAvg	8.0	10.0	9.3	9.0	7.7	7.3	7.3	6.8	6.6	7.1	7.7	8.5	9.1
Precipitation	PrecAvg	30.1	2.7	2.1	1.9	1.6	2.1	2.8	3.1	3.0	2.6	2.4	2.8	3.0
	PrecMax	39.6	6.1	5.7	4.8	3.4	5.5	6.6	8.1	6.7	5.6	7.0	6.7	7.4
	PrecMin	20.7	0.0	0.2	0.2	0.2	0.4	0.1	0.7	1.1	1.0	0.9	0.6	1.0
	PrecStd	4.8	1.5	1.1	1.1	0.8	1.1	1.5	1.7	1.6	1.3	1.1	1.6	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.1	52.9	59.1	70.4	75.4	80.5	86.5	83.7	77.3	66.4	57.3	51.9
		MCWB	50.2	49.9	50.7	57.1	62.5	66.6	68.6	67.7	64.9	61.5	53.8	49.3
	2%	DB	49.3	49.5	55.0	65.4	71.2	76.5	81.8	79.4	73.2	62.6	54.3	49.0
		MCWB	47.2	47.5	48.2	54.6	59.5	65.0	67.0	66.8	63.7	58.5	52.0	46.9
	5%	DB	47.2	46.4	51.7	61.8	67.5	72.4	78.6	75.4	69.8	59.8	52.1	46.5
		MCWB	45.2	44.5	47.8	52.3	58.0	62.7	66.2	65.0	62.1	56.6	50.3	45.0
	10%	DB	45.3	43.7	48.6	58.0	64.4	68.9	74.8	72.5	66.5	57.7	50.2	44.5
		MCWB	43.6	41.8	45.0	51.0	56.6	60.6	64.5	64.1	60.2	55.2	48.5	43.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.4	51.4	52.8	58.2	65.1	69.5	71.6	70.4	66.9	62.0	54.8	50.0
		MCDB	52.4	52.1	56.2	67.9	73.2	76.6	81.0	79.5	74.0	65.2	56.2	51.2
	2%	WB	47.6	47.5	50.0	55.6	61.7	66.8	69.4	67.8	64.6	59.2	52.2	47.1
		MCDB	49.0	49.2	52.6	63.0	67.9	74.2	78.4	75.6	70.5	61.6	53.9	48.6
	5%	WB	45.5	44.7	47.6	53.6	59.2	64.2	67.5	66.4	62.9	57.3	50.4	45.1
		MCDB	46.8	46.1	51.7	60.2	65.0	69.9	75.5	73.0	68.8	59.3	51.9	46.3
	10%	WB	43.8	42.0	45.2	51.6	57.1	61.8	65.7	65.1	61.0	55.5	48.7	43.2
		MCDB	45.3	43.6	48.4	57.2	62.8	67.1	72.7	70.8	65.8	57.4	50.0	44.3
Mean Daily Temperature Range	5% DB	MDBR	7.4	7.9	11.4	15.9	16.8	16.2	16.6	14.7	14.1	10.6	7.6	7.0
		MCDBR	7.7	9.2	17.0	23.2	22.6	22.0	23.8	20.1	19.7	12.6	8.1	8.1
		MCWBR	7.5	8.0	11.8	13.9	13.5	12.3	10.8	10.0	11.1	9.2	7.3	7.8
	5% WB	MCDBR	7.6	9.0	15.1	20.2	18.9	19.3	21.4	16.9	17.9	11.6	8.0	8.0
		MCWBR	7.6	8.0	11.3	12.9	12.5	12.1	11.1	9.3	10.8	9.0	7.5	7.8
Clear-Sky Solar Irradiance	taub	0.318	0.338	0.368	0.409	0.401	0.398	0.417	0.413	0.386	0.358	0.335	0.309	
	taud	2.313	2.325	2.269	2.202	2.289	2.325	2.289	2.314	2.373	2.418	2.368	2.307	
	Ebn at Noon	199	236	254	258	268	270	261	255	247	227	192	177	
	Edh at Noon	18	25	33	40	39	38	39	36	30	23	18	15	
All-Sky Solar Radiation	RadAvg	164	368	768	1297	1645	1736	1627	1324	960	511	210	120	
	RadStd	22	57	74	162	187	124	201	96	100	64	25	8	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-268	N/A	N/A

Nomenclature: See separate page